

# W3WI\_110.2 - Distributed Systems



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**AI Usage:** AI assistants (in particular Claude, but also ChatGPT, Ollama with Gemma/LLama/Qwen, OpenCode with Kimi, etc.) were used to assist in the creation of the slides. This was done, for example, to efficiently generate graphics (i.e. SVG files), or to generate summary tables. Furthermore, AI was used for general quality assurance. Content that may have been suggested by an AI was explicitly validated and adapted whenever it was incorporated.

# Core Topics

- Terminology, concepts, architectures, requirements profiles and architecture models for distributed systems
- Design and implementation approaches
- Comparison of different middleware concepts
- Synchronous and asynchronous communication, remote method calls
- Asynchronous communication and messaging systems
- Security aspects in distributed systems

## Previous Knowledge

- Who is familiar with Java?
- Who is familiar with Python?

# Examination - Portfolio

## Background

- the module *Developing Distributed Systems* has 55 lecture hours
- the lecture **Distributed Systems has 22 lecture hours**
  - ⇒ Distributed Systems will contribute up to **50** points (out of 120) to the final grade for the module. (Please, don't do the math.)

## 2 Parts

**01** Presentations

**02** Programming Exercise

# Presentations - General Conditions

- Each person should present for 10 Minutes sharp!
- The presentations should deal in particular with the core content of the lecture and *be of a conceptual nature*.
  - I. e., briefly describe the underlying problem before you discuss, e.g., the algorithm/technology and how it solves the specific problem, how errors are dealt with, which services are offered, which guarantees/security aspects are implemented or how scalability is achieved.
  - No promotional presentations and no extended motivation presentations. Focus on the topic at hand.
- Presentations have to be in English.

# Presentations - Submissions

## ▲ Attention!

You have to upload your presentation to Moodle up until 7 o'clock on the day of your presentation.

## ▲ Attention!

[Further Requirements and Checklist \(in Ger.\)](#)

The checklist has to be signed and uploaded together with the presentation. (As a second PDF file.)

# Presentations - Available Topics

## ✂ Hint

Students giving presentations belonging to the same block have to coordinate with each other to avoid any overlap. If you need specific knowledge, which is expected to be covered by another student, but you are not sure whether it will be presented sufficiently, create a backup slide for your presentation that covers this topic as well and mark it as a backup slide. This backup slide will not be counted towards the time limit when you give your presentation.

For some topics, we provide some *initial ideas* regarding the potential content of the presentation. These are just suggestions and you are free to deviate from them as long as you cover the core content of the topic and provide a conceptual presentation.

## !! Important

You are also not required to discuss all topics mentioned in the initial ideas. It is your job to determine the key points and provide a coherent and meaningful presentation.

## Virtualization and Virtualization Platforms

### 01 Introduction to Virtualization & Use Cases

- What is virtualization? Historical context and motivation
- Different types/levels of virtualization
- Key use cases: server consolidation, cloud computing, development/testing, isolation, ...
- Benefits and trade-offs

### 02 Hypervisors - Architecture & Types

- What is a hypervisor?
- Type 1 (bare-metal) vs Type 2 (hosted) - architectural differences
- Full virtualization vs paravirtualization approaches
- Examples and when to use each type

### 03 Virtual Machines - Implementation & Management

- VM structure and components (virtual hardware, guest OS)
- VM snapshots
- VM migration (live migration concepts)
- Resource allocation and isolation

### 04 Containers & OS-level Virtualization

- Container concept and how it differs from VMs
- Namespaces and cgroups (conceptual)
- Container images and layering
- Use cases and comparison with VMs

### 05 Memory Virtualization

- The address translation problem (guest virtual → guest physical → host physical)
- Shadow page tables approach
- Hardware-assisted virtualization (EPT/NPT)
- Memory management techniques (overcommitment, ballooning)

## 06 Network & I/O Virtualization

- Challenges of virtualizing network and I/O devices
- Emulated vs paravirtualized devices
- SR-IOV (Single Root I/O Virtualization) and device passthrough
- Virtual NICs and network bridges
- Virtual switches and network isolation

## Network Protocols

01 QUIC (only available when we have  $\geq 21$  students)

02 HTTP/3

03 BitTorrent Protocol (only available when we have  $\geq 25$  students)

## Modern RPC

01 Protobuf

02 Google RPC

## Web-App Security

01 SOP (Same-Origin Policy), CORS (Cross-Origin Resource Sharing) (Foundations)

02 CORP / COOP / COEP (Cross-Origin Resource/Opener/Embedder Policies) (only available when we have  $\geq 23$  students)

03 CSP (Content Security Policy) and SRI (Subresource Integrity)

Introduction and concrete examples how they are used/specified and help prevent attacks.

## Monitoring & Debugging Distributed Systems

01 Log Aggregation with a particular focus on correlation of log entries collected from different nodes in a distributed system.

## Leader Election

01 Bully Algorithm and Ring Algorithm

## Quorum Systems

01 Majority voting (i. e., quorum-distributed computing)

## Consensus Algorithms and Fault Tolerance

01 Consensus Fundamentals & Problem Definition

- What is consensus and why is it hard in distributed systems?
- The FLP impossibility result (conceptual understanding)
- Fault models: crash faults vs Byzantine faults
- Safety vs liveness properties
- Real-world motivation: replicated state machines, distributed databases

02 (Practical) Byzantine Fault Tolerance

- When do we need BFT?

- Modern developments
- Real-world usage

### 03 Paxos Family

- Basic Paxos algorithm (conceptual overview, roles: proposers, acceptors, learners)
- Why Paxos is correct but complex
- Multi-Paxos for practical systems
- Real-world usage

### 04 Raft - Understandable Consensus

- Leader election, log replication, safety
- How Raft differs from Paxos (design philosophy)
- Real-world usage

## Eventual Consistency

### 01 Eventual Consistency and Gossip Protocol

### 02 CRDTs (Conflict-free Replicated Data Types) (only available when we have $\geq 22$ students)

## Distributed File Systems

### 01 Ceph

### 02 HDFS (only available when we have $\geq 24$ students)

# Topic Assignment

The presentations need to be given in the order listed below.

| Topic                                              | Constraint         | Student Name |
|----------------------------------------------------|--------------------|--------------|
| 01. Introduction to Virtualization & Use Cases     |                    |              |
| 02. Hypervisors - Architecture & Types             |                    |              |
| 03. Virtual Machines                               |                    |              |
| 04. Containers & OS-level Virtualization           |                    |              |
| 05. Memory Virtualization                          |                    |              |
| 06. Network & I/O Virtualization                   |                    |              |
| 07. QUIC                                           | $\geq 21$ students |              |
| 08. HTTP/3                                         |                    |              |
| 09. BitTorrent Protocol                            | $\geq 25$ students |              |
| 10. Protobuf                                       |                    |              |
| 11. Google RPC                                     |                    |              |
| 12. SOP, CORS (Foundations)                        |                    |              |
| 13. CORP / COOP / COEP                             | $\geq 23$ students |              |
| 14. CSP and SRI                                    |                    |              |
| 15. Log Aggregation                                |                    |              |
| 16. Leader Election: (a) Bully, (b) Ring Algorithm |                    |              |
| 17. Quorum Systems (Majority Voting)               |                    |              |
| 18. Consensus Fundamentals & Problem Definition    |                    |              |
| 19. Byzantine Fault Tolerance (PBFT)               |                    |              |
| 20. Paxos Family                                   |                    |              |
| 21. Raft - Understandable Consensus                |                    |              |
| 22. Eventual Consistency and Gossip Protocol       |                    |              |
| 23. CRDTs                                          | $\geq 22$ students |              |
| 24. Ceph                                           |                    |              |
|                                                    |                    |              |

|          |                    |  |
|----------|--------------------|--|
| 25. HDFS | $\geq 24$ students |  |
|----------|--------------------|--|

# Presentations - Grading (max 20 Points)

■ Presentation (max 17 Points)

■ Checklist (max 1 Point)

■ Moderation (max 2 Points)

A Moderator has three main tasks:

1. (Introduce the speaker) and *motivate the topic*.
2. Keep track of time and *give time warnings to the presenter* (2 minutes left, time is up); abort the presentation if necessary (-1 minute).
3. Lead a short Q&A session (2-3 questions) after the presentation; if there are no questions from the audience, *the moderator should have two to three questions* that the presenter is not aware of in advance (e.g., by asking the presenter for potential questions related to the topic beforehand).

## Programming Task (max 30 Points)

- Carried out in Groups of two students (and one group with three students if we have an odd number of students)
- You'll get 1 point for a correct upload of the required files.
- You'll get 1 point if the public test suite runs successfully on your solution.
- You'll get up to 8 points based on the results of running an extended test suite on your solution. The grading will be based on the number of test cases passed successfully.
- Grading will be done during an examination interview (20 Points) (15 Minutes per Group):
  1. Demonstration of your project (~5 Minutes)
  2. Answering specific questions related to your implementation
- The download will be made available in due time.

# Lecture - Schedule

**May 18, 2026 (5VL):** ■ Lecture

**June 3, 2026 (2VL):** ■ Lecture

■ Programming Task Explanation

**June 16, 2026 (5VL):** ■ Presentations:

- 01. Introduction to Virtualization & Use Cases (Baha Sahin)
- 02. Hypervisors - Architecture & Types (Thorge Hartmann)
- 03. Virtual Machines (Jan Wendlandt)
- 04. Containers & OS-level Virtualization (Heba Alkhaled)
- 05. Memory Virtualization (Jakob Lindqvist)
- 06. Network & I/O Virtualization (Can Sever)
- 07. QUIC (Simon Wagner)
- 08. HTTP/3 (Yuxi Yan)

■ Lecture

**June 24, 2026 (5VL):** ■ Presentations:

- 10. Protobuf (Mick Burcksen)
- 11. Google RPC (Luisa Homola)
- 12. SOP, CORS (Foundations) (Mia Roth)
- 13. CORP / COOP / COEP (Eliano-Markus Glock)
- 14. CSP and SRI (Giulia Petry)
- 15. Log Aggregation (Omid Saidi)
- 16. Leader Election: Bully and Ring Algorithm (Johanna Jansen)
- 17. Quorum Systems (Majority Voting) (Armin Mirzakhani)

■ Lecture

**30 June, 2026 (5VL):** ■ Presentations:

- 18. Consensus Fundamentals & Problem Definition (Jonathan Lahmer)
- 19. Byzantine Fault Tolerance (PBFT) (Simon John)
- 20. Paxos Family (Janis Josefowski-Pätz)
- 21. Raft - Understandable Consensus (Daria Makaschin)
- 22. Eventual Consistency and Gossip Protocol (Aleksander Orlowski)
- 23. CRDTs (Luca Santoro)
- 24. Ceph (Leni-Marie Sternberger)
- 25. HDFS (Stephan Fürstenau)

■ Lecture

**July 27 2026, 07:00 (7:00 a.m.):**

## ■ Programming Task Submission Deadline

Uploaded by *one team member* only:

- the two JAR files as described in the task description
- a ZIP file with the source code
- a PDF file which explicitly states who worked on which part and which (optionally!) states whether you want to be graded as a group or individually w.r.t the submitted code. The oral exam is always graded individually.

**July 28, 2026:**

## ■ Oral Examination (Programming Task)

### **Duration: 15 minutes**

During the oral examination, you will be asked to present, demonstrate, and explain your solution.

The examination will focus on your solution to the programming task and the concepts directly related to it. It will not cover unrelated topics.

You are required to bring your own notebook computer with a functioning IDE, all source code, and precompiled binaries. Your solution must be fully operational and ready to run on your device.

## Moderation - June 16, 2026 (5VL)

| Topic                                                       | Moderator           |
|-------------------------------------------------------------|---------------------|
| 01. Introduction to Virtualization & Use Cases (Baha Sahin) | Luisa Homola        |
| 02. Hypervisors - Architecture & Types (Thorge Hartmann)    | Mia Roth            |
| 03. Virtual Machines (Jan Wendlandt)                        | Armin Mirzakhanian  |
| 04. Containers & OS-level Virtualization (Heba Alkhaled)    | Johanna Jansen      |
| 05. Memory Virtualization (Jakob Lindqvist)                 | Eliano-Markus Glock |
| 06. Network & I/O Virtualization (Can Sever)                | Omid Saidi          |
| 07. QUIC (Simon Wagner)                                     | Giulia Petry        |
| 08. HTTP/3 (Yuxi Yan)                                       | Mick Burcksen       |

## Moderation - June 24, 2026 (5VL)

| Topic                                                          | Moderator              |
|----------------------------------------------------------------|------------------------|
| 10. Protobuf (Mick Burcksen)                                   | Janis Josefowski-Pätz  |
| 11. Google RPC (Luisa Homola)                                  | Leni-Marie Sternberger |
| 12. SOP, CORS (Foundations) (Mia Roth)                         | Simon John             |
| 13. CORP / COOP / COEP (Eliano-Markus Glock)                   | Luca Santoro           |
| 14. CSP and SRI (Giulia Petry)                                 | Daria Makaschin        |
| 15. Log Aggregation (Omid Saidi)                               | Aleksander Orlowski    |
| 16. Leader Election: Bully and Ring Algorithm (Johanna Jansen) | Stephan Fürstenau      |
| 17. Quorum Systems (Majority Voting) (Armin Mirzakhanian)      | Jonathan Lahmer        |

## Moderation - 30 June, 2026 (5VL)

| Topic                                                              | Moderator       |
|--------------------------------------------------------------------|-----------------|
| 18. Consensus Fundamentals & Problem Definition (Jonathan Lahmer)  | Baha Sahin      |
| 19. Byzantine Fault Tolerance (PBFT) (Simon John)                  | Can Sever       |
| 20. Paxos Family (Janis Josefowski-Pätz)                           | Jakob Lindqvist |
| 21. Raft - Understandable Consensus (Daria Makaschin)              | Yuxi Yan        |
| 22. Eventual Consistency and Gossip Protocol (Aleksander Orlowski) | Simon Wagner    |
| 23. CRDTs (Luca Santoro)                                           | Jan Wendlandt   |
| 24. Ceph (Leni-Marie Sternberger)                                  | Heba Alkhaled   |
| 25. HDFS (Stephan Fürstenau)                                       | Thorge Hartmann |

## Exam Interviews (July, 28 2026)

|       |                     |                        |
|-------|---------------------|------------------------|
| 13:00 | Jonathan Lahmer     | Thorge Hartmann        |
| 13:20 | Johanna Jansen      | Luisa Homola           |
| 13:40 | Armin Mirzakhanian  | Heba Alkhaled          |
| 14:00 | Yuxi Yan            | Leni-Marie Sternberger |
| 14:20 | Pause               | Pause                  |
| 14:30 | Daria Makaschin     | Stephan Fürstenau      |
| 14:50 | Mia Roth            | Giulia Petry           |
| 15:10 | Baha Sahin          | Simon John             |
| 15:30 | Luca Santoro        | Can Sever              |
| 15:50 | Pause               | Pause                  |
| 16:00 | Simon Wagner        | Janis Josefowski-Pätz  |
| 16:20 | Aleksander Olowski  | Omid Saidi             |
| 16:40 | Eliano-Markus Glock | Jakob Lindqvist        |
| 17:00 | Jan Wendlandt       | Mick Burcksen          |